4	(a)	magnitudes of <u>maximum</u> displacements / accelerations are different graph is curved / not a straight line (at both ends of graph)	B1 B1
	(b) (i)	Simple harmonic motion so $a = -\omega^2 x$ where ω is the angular frequency $\omega^2 = (2)(64)/(0.810)$ $\omega = 12.57$ rad s^{-1} $f = \omega / 2\pi$ $= 2.0$ Hz	B1 B1
	(b)(ii)	max speed $= \omega x_o$ $= (12.57)(0.016)$ $= 0.20$ m s^{-1}	 C1 A1
	(iii)	momentum is conserved (horizontally), $m_{trolley}\ v_{trolley} = m_{total}\ v_{max}$ max speed decreases Or perfectly inelastic collision, energy is lost, so max speed decreases	M1 A1
		perfectly inelastic collision so energy lost less elastic potential energy so amplitude decreases	M1 A1